

Todor Hristov

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Education

- Sep 2021 – **Computer Science (MSc)**, *ETH Zürich*, Switzerland
March 2025
 - Overall Grade: 5.38/6.00
 - Received the Jury Award in the Game Programming Laboratory course.
 - Units studied include:
 - Advanced Systems Lab
 - Information Security Lab
 - Program Verification
 - Automated Software Testing
 - System Security
 - Principles of Distributed Computing
 - Design of Parallel and High-Performance Computing
- Sep 2017 – **Computer Science (BSc)**, *University of Bristol*, UK
July 2020
 - Recipient of The Hargrave Scholarship for Academic Achievement**
 - Overall Grade: 81% (First Class Honours)
 - Attended the Algorithms Research Reading Group.
 - Units studied include:
 - Machine Learning
 - Computational Neuroscience
 - Data Structures and Algorithms
 - Concurrent Computing
 - Object-Oriented Programming
 - Software Product Engineering
 - An Introduction to High Performance Computing
- Sep 2012 – **Secondary Education**, *High School of Mathematics "Dr Petar Beron"*, Bulgaria
June 2017
 - Overall Grade: 5.98/6.00 | Major: Informatics and Mathematics

Work Experience

- June 2025 – **Software Engineer** → **Computer Architecture Engineer**, *Huawei*, Zürich, Switzerland
Present
 - Computer Architecture Engineer (March 2026 – Present)**
 - Working on processor microarchitecture design and performance optimisation.
 - Collaborating with cross-functional hardware and software teams on CPU development.
 - Gaining experience in computer architecture and low-level hardware design.
 - Software Engineer (June 2025 – February 2026)**
 - Added Model Context Protocol (MCP) support to the LLM-enabled CodeArts IDE.
 - Designed and optimised algorithms that reduced the overall token usage by 43% to 52%.
 - Validated the MCP functionality using MCP-Bench (github.com/Accenture/mcp-bench).
- March 2023 – **Research Assistant**, *Oracle*, Zürich, Switzerland
August 2023
 - Contributed to the development of the Parallel Graph AnalytiX (PGX) platform.
 - Designed and optimised algorithms for efficient computation on large, complex graphs.
 - Implemented new features and enhancements for the PGX platform, leveraging Java and Python.
- Sep 2020 – **Backend Developer**, *Inexogy Smart Metering (Discovergy GmbH)*, Heidelberg, Germany
August 2021
 - Developed a Smart Meter Gateway Administrator (SMGWA) for energy data management.
 - Used Go for its development, optimising system performance and integration between components.
 - Leveraged Bash for deployment and automation, streamlining the different processes.
 - Gained hands-on experience in the full development life cycle.
- June 2018 – **Systems Analyst Intern**, *VarSys Ltd.*, Varna, Bulgaria
August 2018
 - Diagnosed and repaired hardware and software issues, improving technical troubleshooting skills.
 - Effectively communicated technical problems to non-technical clients.
 - Researched Ethereum mining hardware and software for a client project as the primary lead.

Projects

- 2024 **Large Language Models for Email Phishing Training and Education**,
Prof. Dr. Srdjan Čapkun, Dr. Daniele Lain, ETH Zürich, Switzerland
- Master's Thesis with the System Security Group, focused on AI-enhanced email phishing training.
 - Built a modular platform with `React`, `FastAPI`, and `MongoDB`, deployed in three `Docker` containers.
 - Conducted a 201-participant study showing that LLMs enhanced training-page effectiveness.
 - Available on ETH Zürich's Research Collection: doi.org/10.3929/ethz-b-000695282
- 2022 **Seth's Pyramall**, *Game Programming Laboratory, ETH Zürich, Switzerland*
- Developed a 2D platformer in `C#` using the `MonoGame` framework, as part of a six-person team.
 - Recipient of the Jury Award, given by the expert game developers from Studio Gobo.
 - Implemented a grid-based sand simulation, creating realistic flow and piling dynamics.
 - YouTube Trailer: youtube.com/watch?v=u3z3P-rHndg
- 2020 **Improving Mouse Detection via Motion Tracking**,
Dr. Sion Hannuna, Dr. Neill Campbell, University of Bristol, UK
- Bachelor's Thesis focused on improving object detection through motion tracking.
 - Applied SORT (tracking-by-detection) to reduce false positives and identify hard-to-detect frames.
 - Performance evaluated against bootstrapping and classical data augmentation methods.
- 2020 **C++ Raytracer**, *Computer Graphics, University of Bristol, UK*
- Developed a raytracer using Simple DirectMedia Layer (SDL) and OpenGL Mathematics (GLM).
 - As part of a two-person team, implemented the following features:
 - Soft shadows
 - Anti-aliasing
 - Mirrors
 - Textures
 - Phong Shading
 - Glass

Achievements

- 2020 **Hargrave Scholarship for Academic Achievement**, *University of Bristol, UK*
- Awarded to the top final-year BSc Computer Science student for outstanding academic performance.
- 2018 – 2019 **Top 100 in the World**, *Bloomberg Global CodeCon Finals, UK*
- Represented the University of Bristol at Bloomberg's global coding finals in 2018 and 2019.
 - Ranked Top 100 worldwide, solving eight advanced algorithmic problems in `C++`.
- 2018 **First Place from University of Bristol**, *AI Gaming Mini-Hack, UK*
- Ranked first among University of Bristol participants in an AI game-agent competition.
 - Developed a `Python`-based approximation to the NP-hard Travelling Salesman Problem.
- 2017 **First Place**, *Accessibility Hack 2017, University of Bristol, UK*
- Awarded first place in a group competition focused on creating solutions to enhance accessibility.
 - Developed a Chrome Extension that detects and simplifies complex website text.
 - GitHub Repository: github.com/LukeStorry/simple_read
- 2015 **Gold Medal**, *National Spring Competition in Informatics, Bulgaria*
- Gold medal (1st place) in one of Bulgaria's most prestigious national programming competitions.
 - Solved three advanced `C++` problems under strict time and memory constraints.

Skills

Technical	Programming Languages:	C/C++, Python, TypeScript, Java, Go, Bash
	Tools & Technologies:	Git, Docker, MPI, $\text{\LaTeX}$
	Software Development:	Agile Software Development, Test-Driven Development
	Hardware Simulation:	Trace-Driven Hardware Simulation (<code>ChampSim</code>)
	Computer Science Fundamentals:	Computer Architecture, Data Structures and Algorithms
Languages	English (C1) German (A2) Bulgarian (Native)	

Hobbies

Playing table football and simulation video games, such as *Cities: Skylines* and *Minecraft*.